

Assignment Questions

1. List Comprehension (Basics)

1. Given the list

```
nums = [2, 5, 7, 10, 13]
```

Create a new list containing the **cube** of each number using list comprehension.

2. From the list

```
nums = [3, 8, 12, 15, 18, 21]
```

Create a new list that contains **only the numbers divisible by 3**.

3. Given

```
words = ["apple", "cat", "banana", "car", "dog"]
```

Create a list that contains only the words **that do NOT start with 'c'**.

2. Dictionary Comprehension

4. Given the dictionary

```
scores = {1: 55, 2: 80, 3: 40, 4: 90}
```

Create a new dictionary containing **only the items where the score is above 60**.

5. Convert this dictionary into a reverse dictionary (swap key ↔ value)

```
colors = {  
    1: "red",  
    2: "blue",  
}
```

```
3: "green"  
}
```

3. Exception Handling

6. Write a program that:

- Takes a number from the user
- Tries to convert it to an integer
- If the user enters something invalid, print:
`"Invalid input! Please enter a number."`

7. Write a program that:

- Asks the user for two numbers
 - Tries to divide them
 - Handles:
 - `ValueError` → when input is not a number
 - `ZeroDivisionError` → when denominator is 0
 - `Exception` → any other error
 - Prints `"Division successful!"` inside the `else` block.
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4. File Handling

File Handling Examples (with Code)

1. Writing to a File (`w` mode)

- ✓ Creates the file if it doesn't exist
- ✓ Overwrites the file if it already exists

Example 1: Write a single line

```
file = open("hello.txt", "w")
file.write("Hello students!")
file.close()
```

Example 2: Write multiple lines

```
file = open("fruits.txt", "w")
file.write("Apple\n")
file.write("Banana\n")
file.write("Mango\n")
file.close()
```

2. Reading from a File (**r** mode)

- ✓ File must already exist
- ✓ Used to read data

Example 3: Read entire file

```
file = open("hello.txt", "r")
content = file.read()
print(content)
file.close()
```

Example 4: Read line by line

```
file = open("fruits.txt", "r")
for line in file:
    print(line.strip()) # strip() removes \n
file.close()
```

Example 5: Read all lines into a list

```
file = open("fruits.txt", "r")
lines = file.readlines()
print(lines)
file.close()
```

3. Appending to a File (**a** mode)

- ✓ Adds new content at the end
- ✓ Does not delete previous data

Example 6: Append lines

```
file = open("fruits.txt", "a")
file.write("Orange\n")
file.close()
```

4. Using **with open()** (Best Practice)

- ✓ Automatically closes the file
- ✓ Prevents mistakes like forgetting `close()`

Example 7: Writing with **with**

```
with open("city.txt", "w") as f:
    f.write("Mumbai\n")
    f.write("Delhi\n")
    f.write("Chennai\n")
```

Example 8: Reading with **with**

```
with open("city.txt", "r") as f:
    print(f.read())
```

5. Example: Reading Numbers & Doing Something with Them

Assume **numbers.txt** contains:

```
10
20
30
40
50
```

Example 9: Read numbers and calculate sum

```
with open("numbers.txt", "r") as f:
    total = 0
    for line in f:
        total += int(line.strip())

print("Total =", total)
```

6. Example: Write User Input to a File

Example 10: Ask the user for 3 names and save them

```
with open("names.txt", "w") as f:
    for i in range(3):
        name = input("Enter a name: ")
        f.write(name + "\n")
```

7. Example: Read a File Safely (with Exception Handling)

Example 11: Handle file-not-found error

```
try:
    with open("unknown.txt", "r") as f:
        print(f.read())
except FileNotFoundError:
    print("File does not exist!")
```
